

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2023

Course: B. Tech. Branch : Civil & Allied Engineering Semester : IV

Subject Name: Water Resources Engineering

Subject Code : BTCVC404/BTCIC404/BTCEC404

Max Marks: 60

Date: 23/1/2024

Duration: 3 Hr.

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) An irrigation canal has gross command area of 90000 hectares out of which 80% is culturable irrigable. The intensity of irrigation of Kharif is 40% and for Rabi season is 70%. Find the discharge required at the head of the canal if the duty at its head is 800 hectares/cumec for Kharif season and 1700 hectares/cumec for Rabi season.	CO1	6
B) Write a short note on crop season and crop pattern.	CO1	6
C) Explain briefly about advantages and ill-effects of irrigation.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Write short note on the selection criterion of site for reservoir.	CO2	6
B) Explain the steps followed to calculate of reservoir capacity for a specific yield from mass inflow curve. Explain with proper diagram.	CO2	6
C) Explain the various classifications of reservoirs.	CO2	6
Q. 3 Solve Any Two of the following.		12
A) Explain briefly the causes of failure of gravity dam.	CO2	6
B) What is a spillway? What are the various types of spillways?	CO2	6
C) The peak of flood hydrograph due to 3h duration storm in a catchment is 250 cumecs. The total depth of rainfall is 5.6cm. Assuming an average infiltration loss of 0.2 cm/h and a constant base flow of 15 cumecs.	CO3	6

1. Estimate the peak of the 3h-UH of this catchment.
2. If the area of this catchment is 500 sq.km., determine the base width of the 3h-UH by assuming it to be triangular in shape.

Q.4 Solve Any Two of the following. 12

- A) Enlist the various types of cross drainage works and explain three of them with proper diagram. **CO2 6**
- B) Explain the three methods of base flow separation from flood hydrograph with proper diagram. **CO3 6**
- C) Why is the lining of the canal necessary? **CO2 6**

Q. 5 Solve Any Two of the following. 12

- A) Explain the curative measures of reclamation of water logged land. **CO1 6**
- B) A catchment area has 7 raingauge stations. In a year the annual rainfall recorded by the gauges are as follows. For a 5% error in the estimation of mean rainfall, calculate the minimum number of additional stations required to be established in the catchment. **CO3 6**

Raingauge Station	A	B	C	D	E	F	G
Rainfall (cm)	130	142	118	108	165	102	146

- C) Compare between lift and canal irrigation. **CO2 6**

*** End ***